Derivation of the Average value of $r_2(n)$

The average of $r_2(n)$ is given by,

$$A(N) = \frac{1}{N} \sum_{n=1}^{N} r_2(n)$$

Each lattice point satisfying:

$$x^2 + y^2 = k$$

for some $k \leq N$ contributes exactly 1 to $r_2(k)$.

Therefore,

$$\sum_{n=1}^{N} r_2(n)$$

counts all lattice points with:

$$x^2 + y^2 \leq N$$

Using Gauss circle estimate,

$$G(r) = \pi r^2 + E(r)$$

with $r = \sqrt{N}$, we have

$$\sum_{n=1}^{N} r_2(n) = \pi N + E(\sqrt{N})$$

Thus,

$$A(N) = \pi + \frac{E(\sqrt{N})}{N}$$

As $N \to \infty$,

$\frac{E(\sqrt{N})}{N} \to 0$ and $A(N) \to \pi$